WEEK 4 – ASSIGNMENT

Title:

Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

Objective:

To understand Azure cloud concepts and build a complete data pipeline using Storage Account and Azure Data Factory.

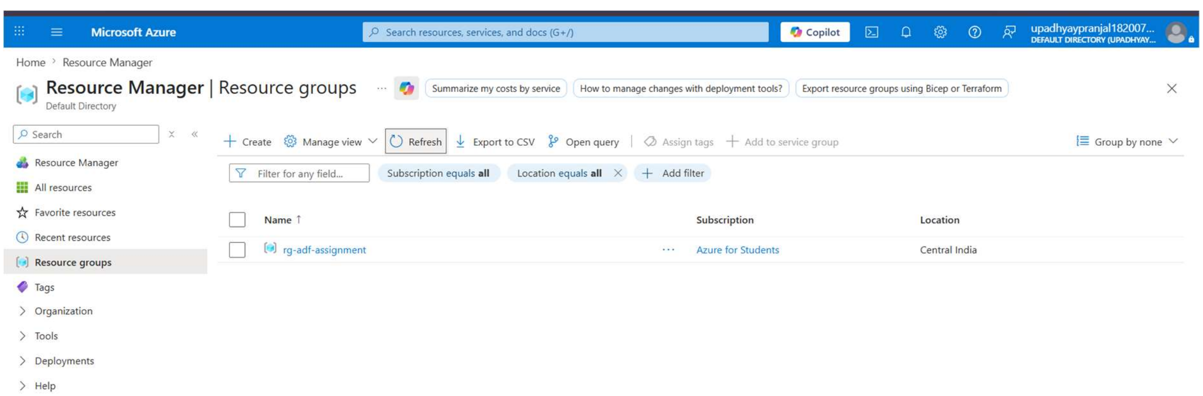
Assignment Tasks

Task 1:

- Explore Azure Portal
- Create a Resource Group

Deliverable:

- Screenshot of Resource Group



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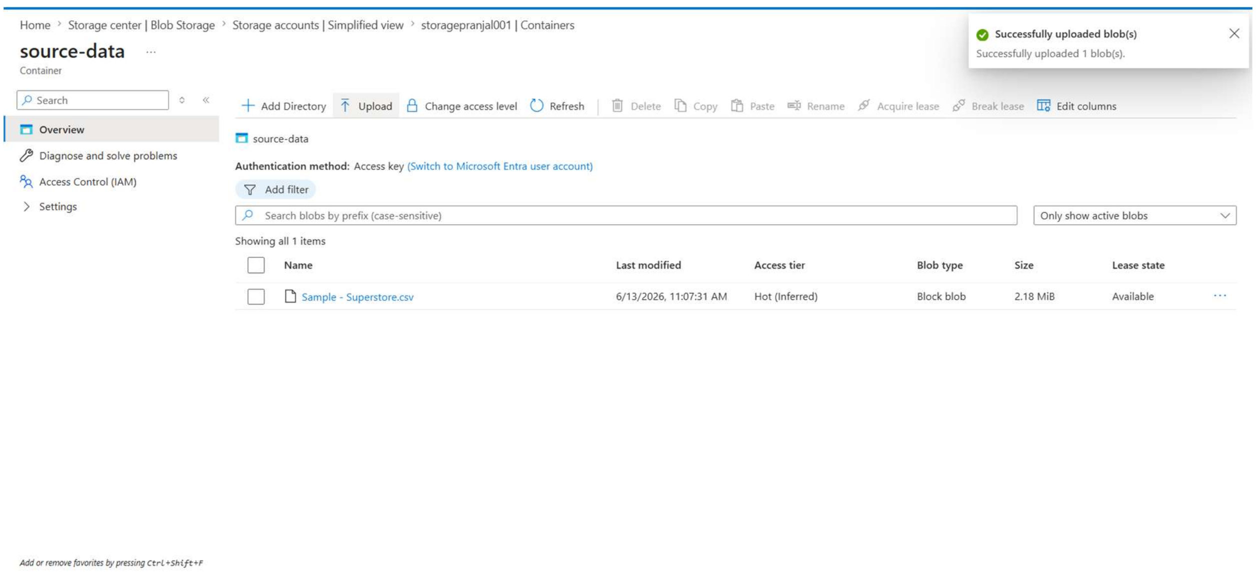
Task 2: Storage Setup

Do:

- Create a Storage Account
- Create a Blob Container
- Upload a CSV file

Deliverable:

- Screenshot of container with uploaded file



Task 3: ADF Basics

Do:

- Create Azure Data Factory
- Explore ADF UI (Author, Monitor, Manage)
- Create Linked Service (Blob Storage)

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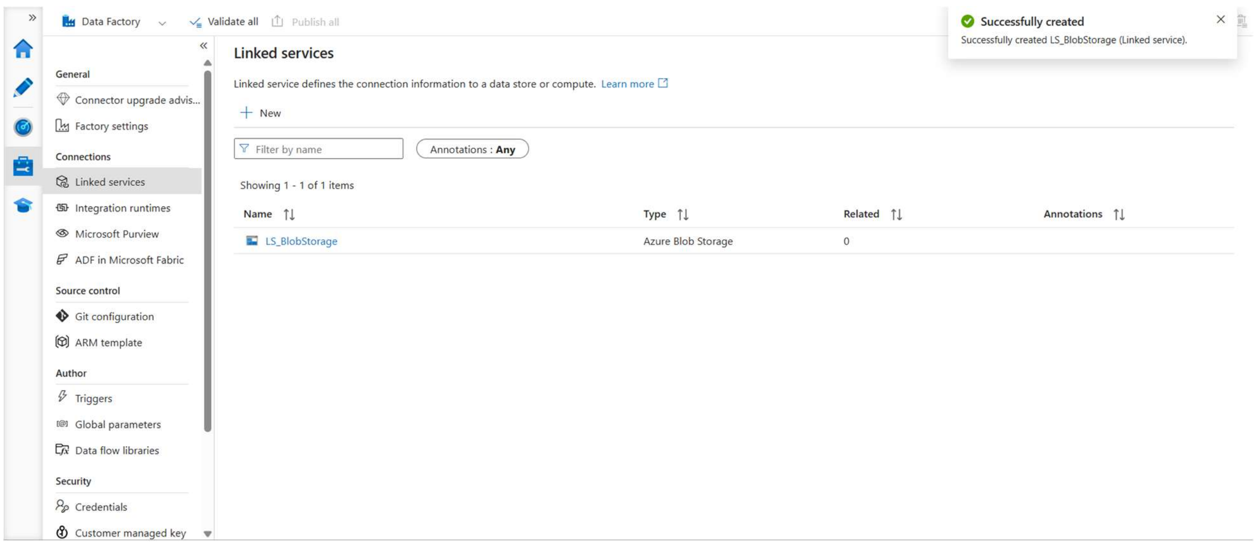
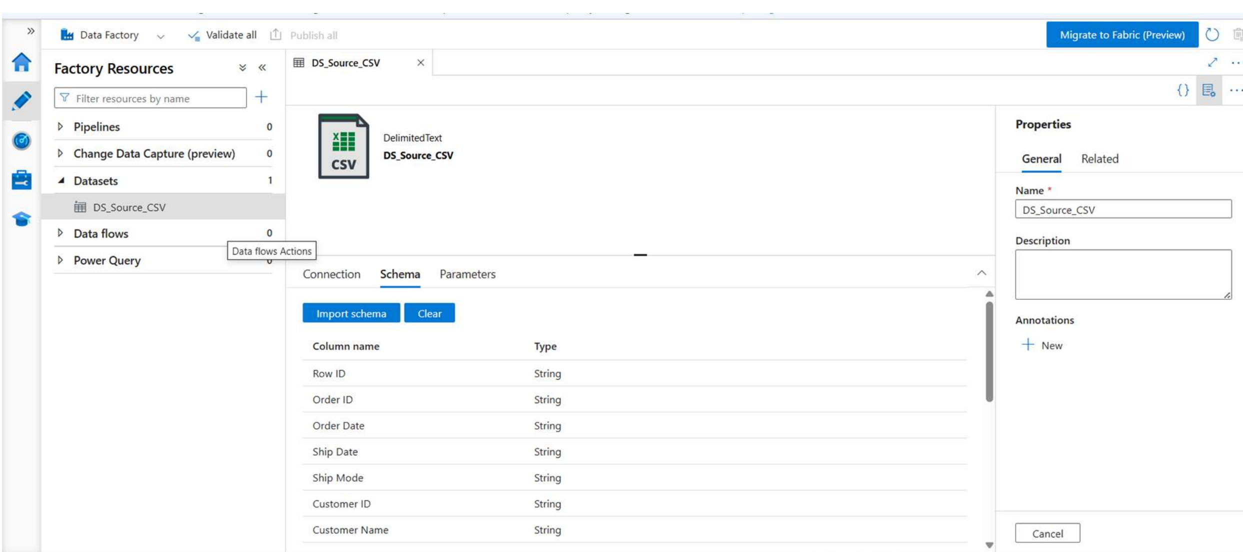
- Create datasets (source + destination)
- Use Get Metadata activity

Deliverable:

- Screenshot of Linked Service
- Screenshot of dataset
- Screenshot of Get Metadata activity





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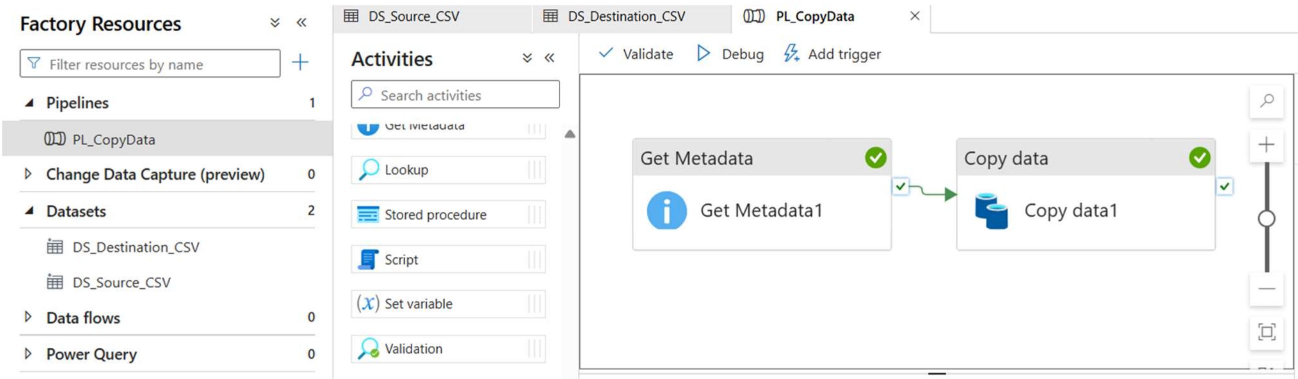


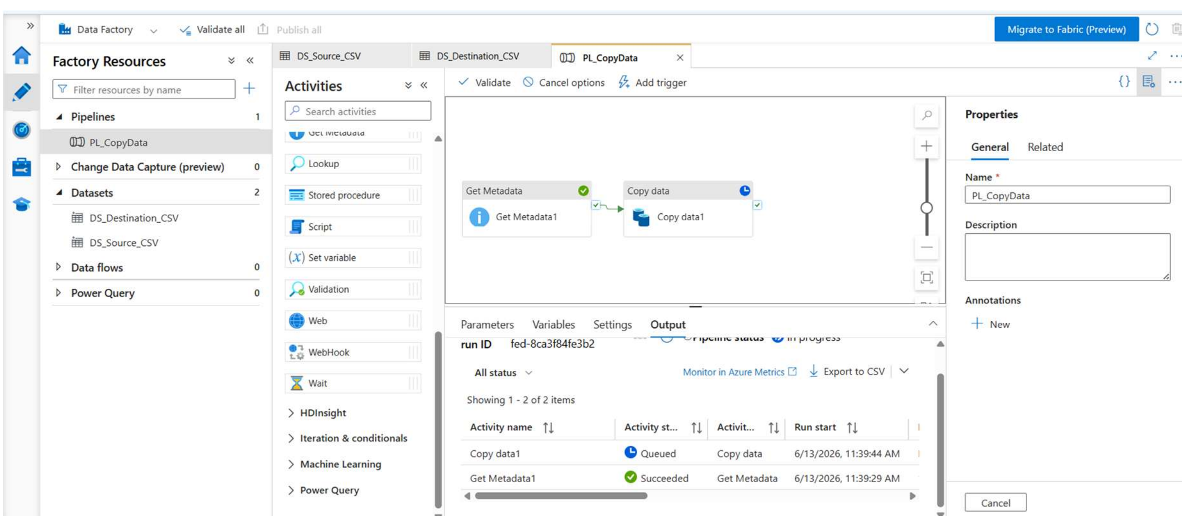
Task 4: Pipeline Development

Do:

- Create pipeline using Copy Data activity
- Configure source and destination
- Add ForEach activity (optional if multiple files)

Deliverable:

- Screenshot of pipeline design



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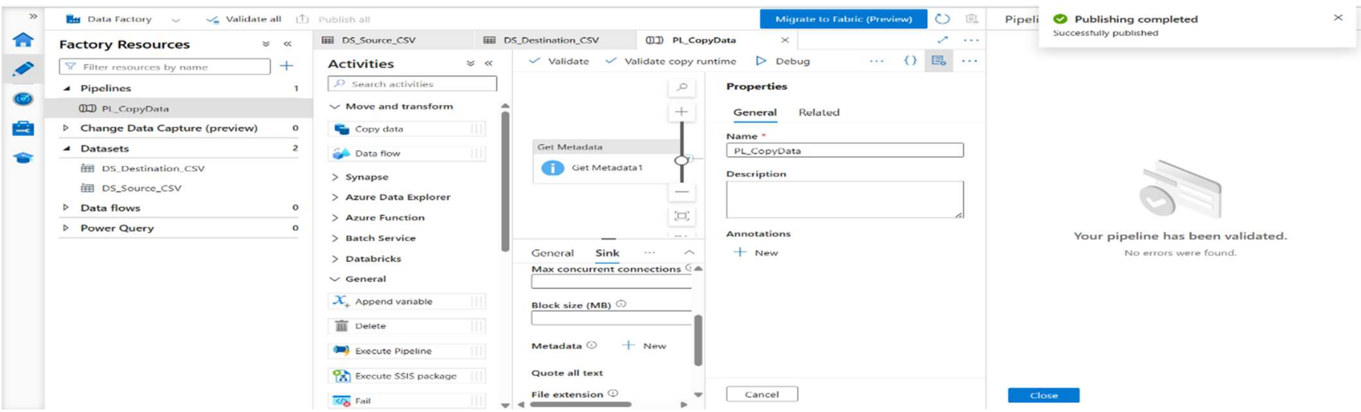
Task 5: Pipeline Execution

Do:

- Run pipeline using Debug/Trigger

Deliverable:

- Screenshot showing pipeline execution (Succeeded)



- Assign roles:
  - Reader
  - Contributor
- Provide access to ADF for Storage

Deliverable:

- Screenshot of role assignment

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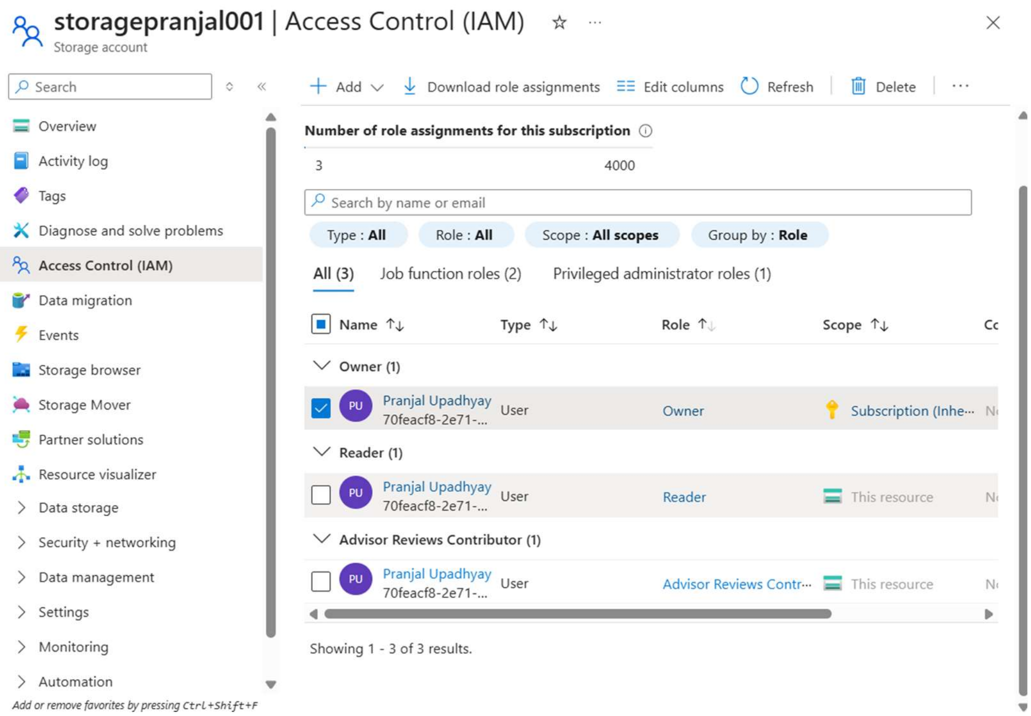

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<u>Mini Project</u>

Title

CSV Data Processing Pipeline using Azure Blob Storage and Azure Data Factory

Problem Statement

Build a complete pipeline that reads a CSV file from Azure Blob Storage and processes it using Azure Data Factory.

Objective

The objective of this project is to design and implement an end-to-end data pipeline using Azure services. The pipeline reads a CSV file stored in Azure Blob Storage, validates the file metadata, and copies the data to a new destination location using Azure Data Factory.

Requirements

Source

CSV file stored in Azure Blob Storage.

Components Used

- Azure Resource Group
- Azure Storage Account
- Azure Blob Containers
- Azure Data Factory
- Linked Service
- Source Dataset
- Destination Dataset
- Get Metadata Activity
- Copy Data Activity

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Process

1. Read CSV file from source container.
2. Validate file metadata using Get Metadata activity.
3. Copy file to destination container using Copy Data activity.
4. Monitor pipeline execution.

Destination

New file location inside Azure Blob Storage.

Solution Architecture

Sample - Superstore.csv

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Azure Blob Storage

(source-data)

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Linked Service

(LS_BlobStorage)

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Source Dataset

(DS_Source_CSV)

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Get Metadata Activity

(File Validation)

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Copy Data Activity

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Destination Dataset

(DS_Destination_CSV)

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output.csv

(destination-data)

Implementation Steps

Step 1: Create Resource Group

A Resource Group named **rg-adf-assignment** was created in Azure Portal to manage all project resources.

Deliverable:

• Screenshot of Resource Group Creation

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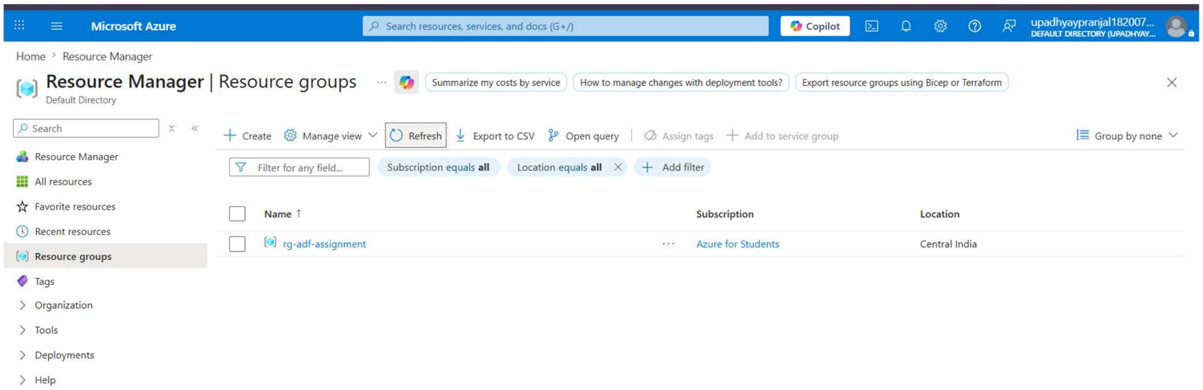


Step 2: Create Storage Account

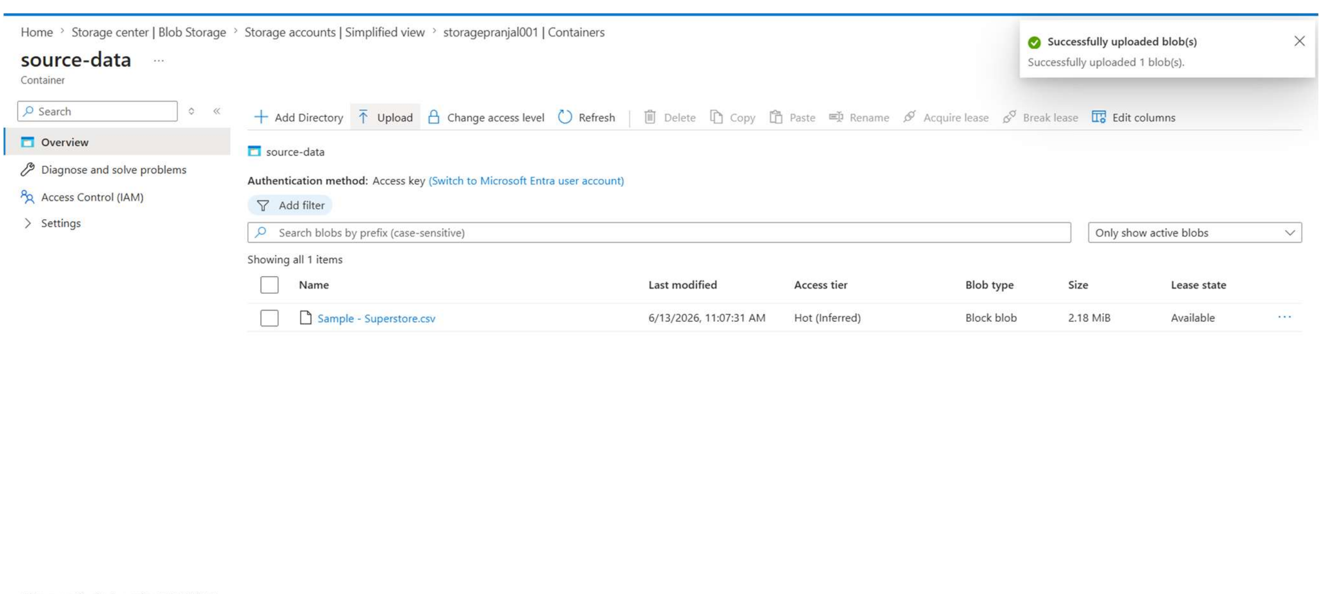
A Storage Account named **storagepranjal001** was created.

Two Blob Containers were created:

- source-data
- destination-data

Deliverable:

• Screenshot of Storage Account



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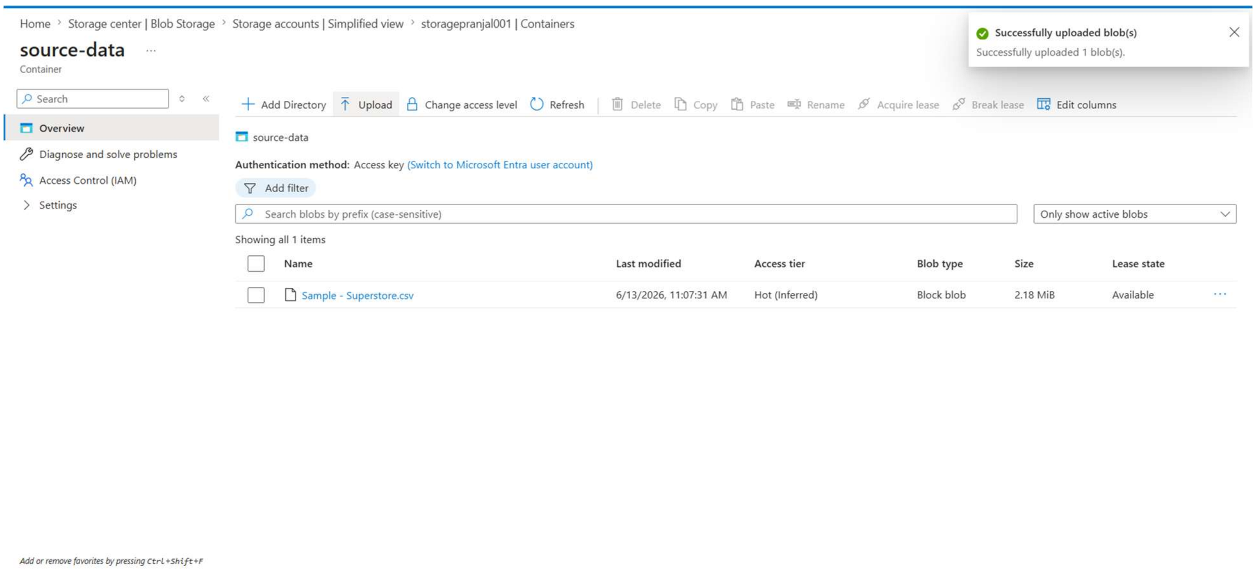
Step 3: Upload CSV File

The file **Sample - Superstore.csv** was uploaded to the **source-data** container.

Deliverable:

• Screenshot showing uploaded CSV file in source-data container



Step 4: Create Azure Data Factory

An Azure Data Factory instance named **adf-pranjal18** was created.

The ADF Studio interface was explored including:

- Author
- Monitor
- Manage

Deliverable:

• Screenshot of Azure Data Factory Home Page

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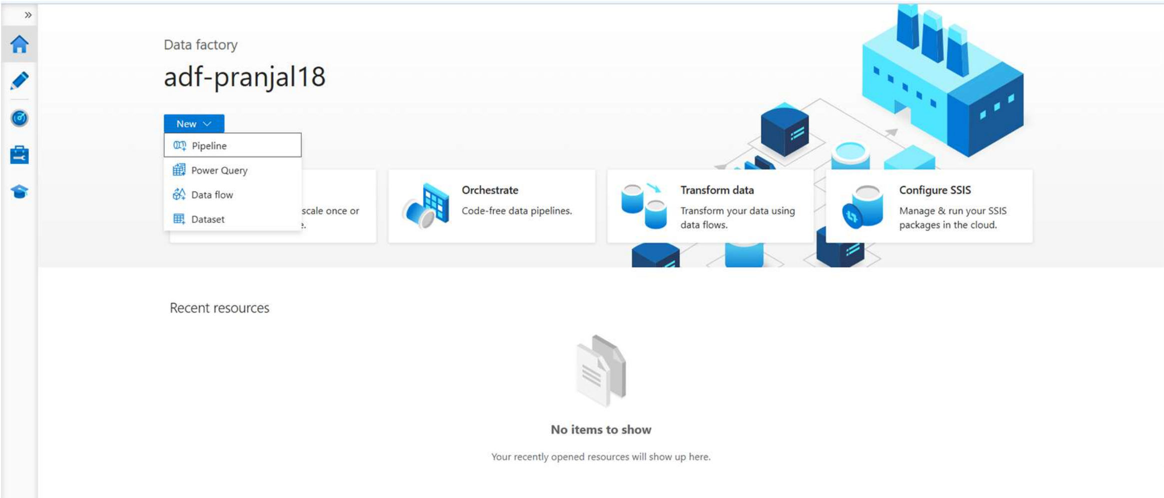
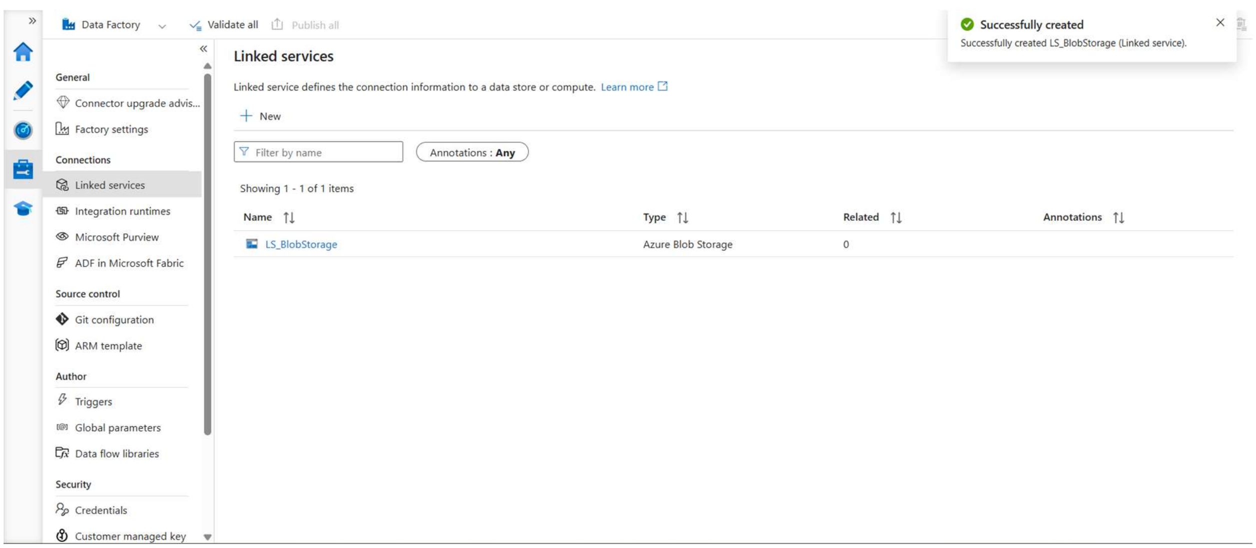


Step 5: Create Linked Service

A Linked Service named **LS_BlobStorage** was created to establish a connection between Azure Data Factory and Azure Blob Storage.

Deliverable:

• Screenshot of Linked Service Configuration



Step 6: Create Source Dataset

Dataset Name:

DS_Source_CSV

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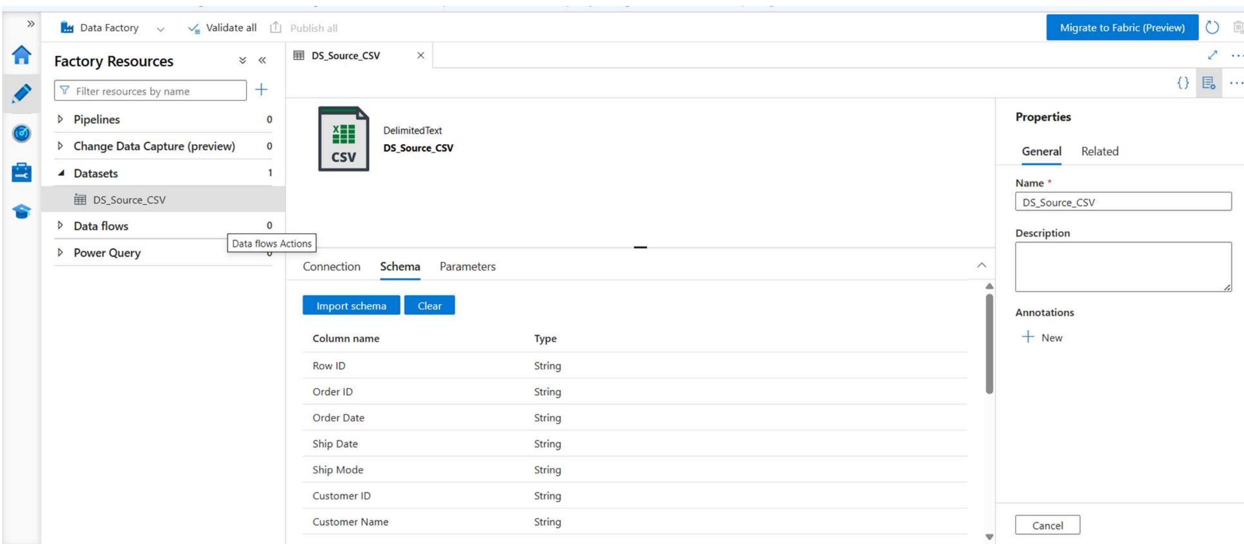
Configuration:

- Linked Service: LS_BlobStorage
- Container: source-data
- File: Sample - Superstore.csv

Deliverable:

• Screenshot of Source Dataset (DS_Source_CSV)



Step 7: Create Destination Dataset

Dataset Name:

DS_Destination_CSV

Configuration:

- Linked Service: LS_BlobStorage
- Container: destination-data
- Output File: output.csv

Deliverable:

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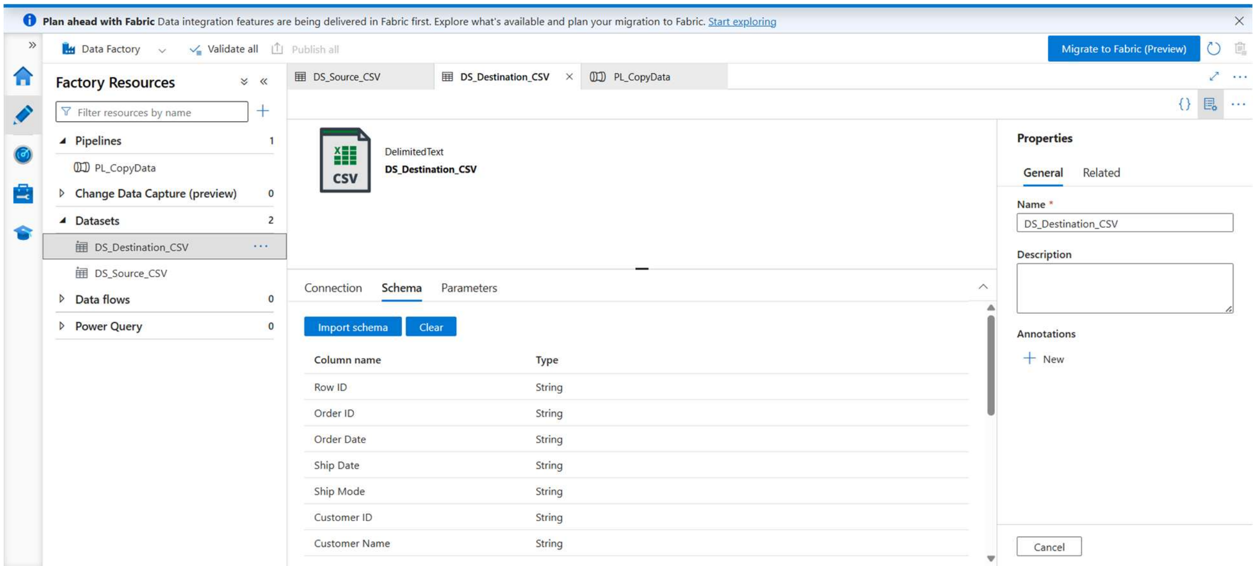
• Screenshot of Destination Dataset (DS_Destination_CSV)



Step 8: Configure Metadata Validation

A **Get Metadata** activity was added to validate the source file before processing.

Metadata Fields Checked:

- Exists
- Size
- Last Modified

Purpose:

To ensure that the source file is available and contains valid metadata before copying.

Deliverable:

• Screenshot of Get Metadata Activity Configuration

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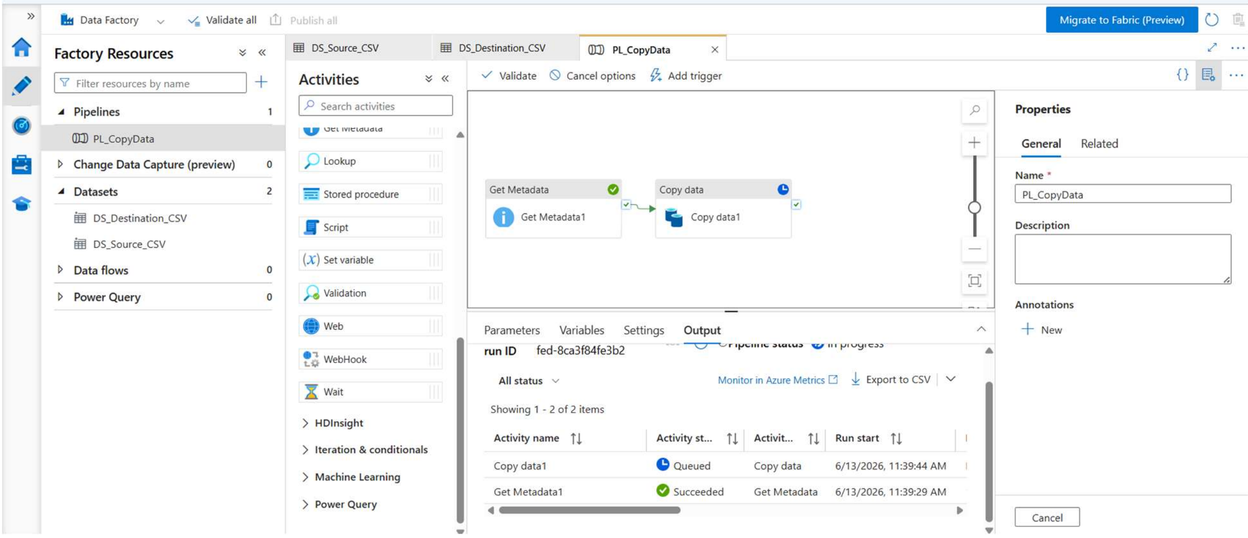

Step 9: Configure Copy Data Activity

A Copy Data activity was configured to copy the source CSV file to the destination container.

Source Dataset:

DS_Source_CSV

Destination Dataset:

DS_Destination_CSV

Deliverable:

• Screenshot of Copy Data Activity Configuration

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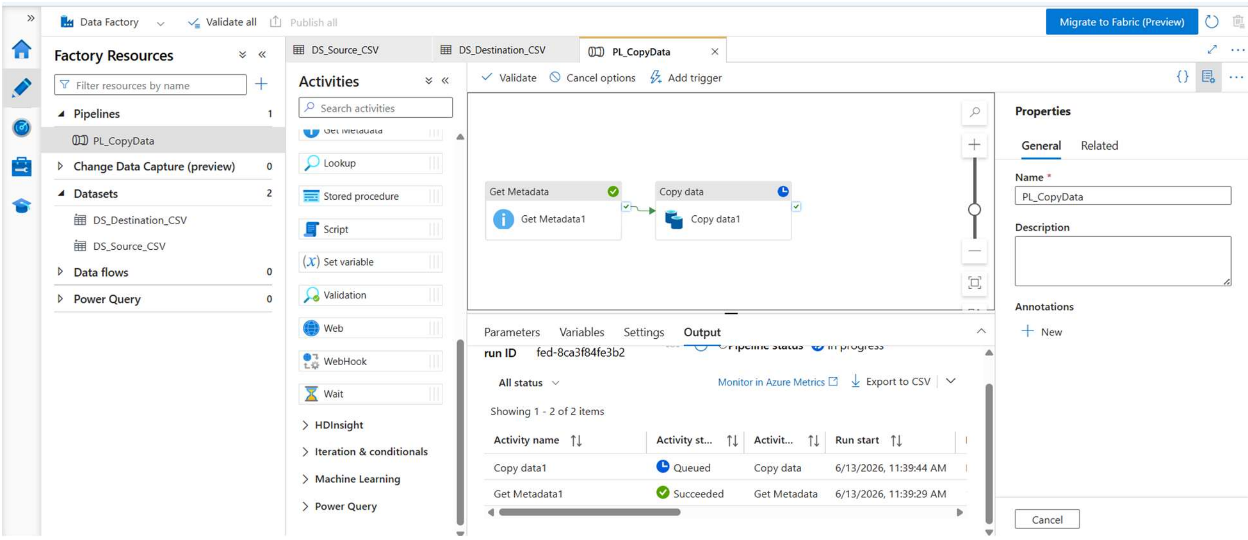

Step 10: Create Pipeline

Pipeline Name:

PL_CopyData

Pipeline Workflow:

Get Metadata

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Copy Data

Deliverable:

• Screenshot of Complete Pipeline Design

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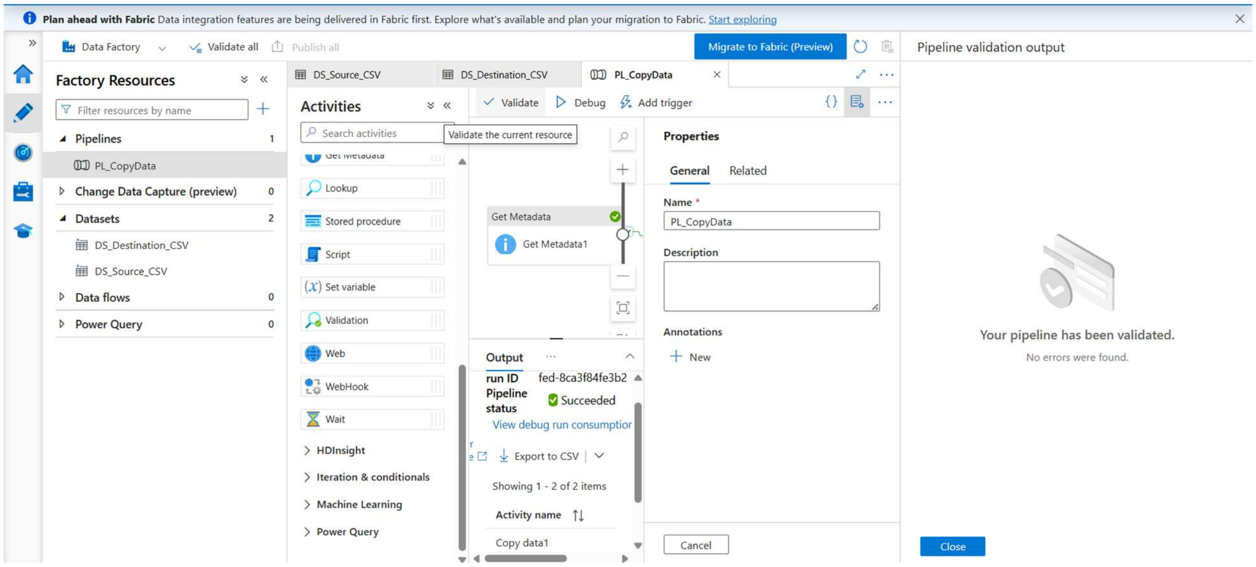
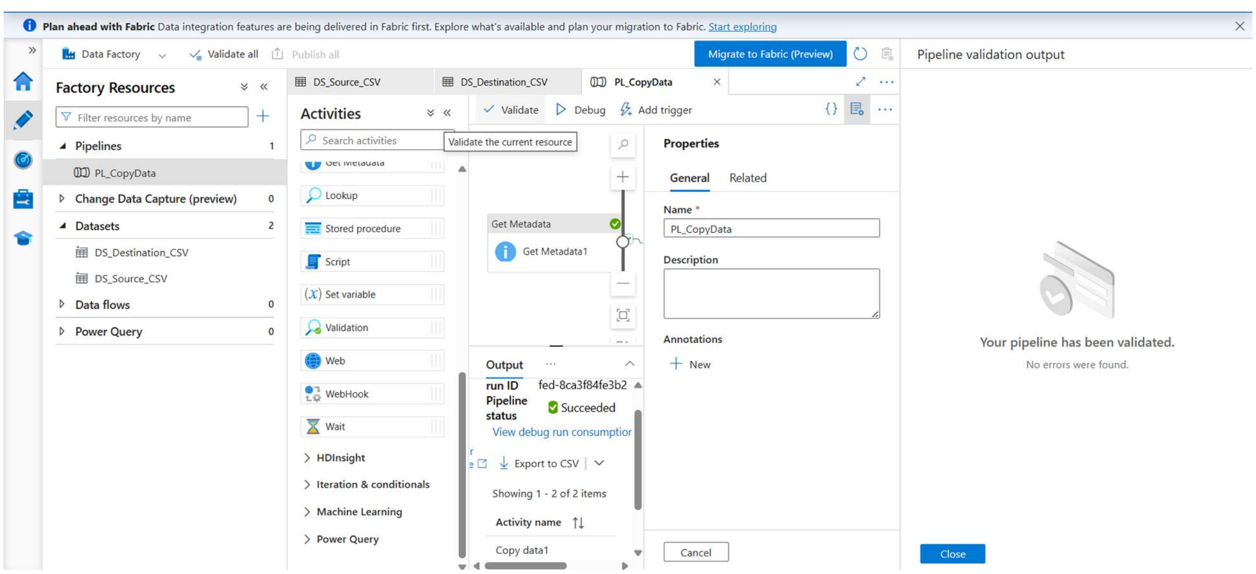


Step 11: Validate and Publish Pipeline

The pipeline was validated successfully and then published to Azure Data Factory.

Deliverable:

• Screenshot showing Validation Successful



Step 12: Execute Pipeline

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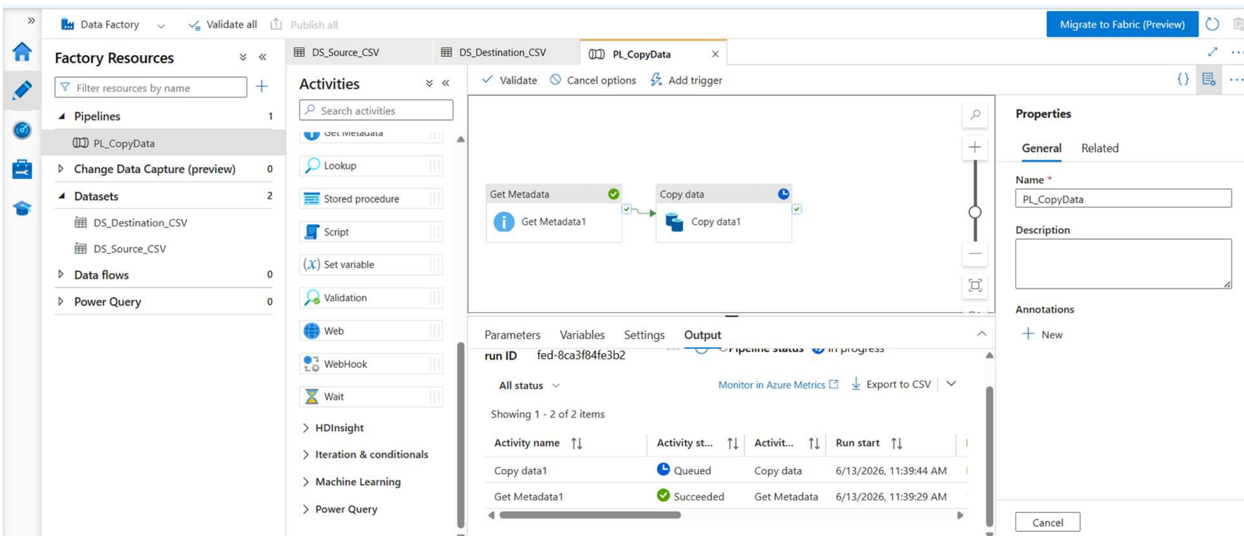
The pipeline was executed using Debug mode.

Execution Process:

- Metadata validation completed successfully.
- Source file processed successfully.
- Data copied to destination container.

Deliverable:

• Screenshot showing Pipeline Execution



Step 13: Verify Output File

After successful execution, the output file was generated in the destination location.

Output File:

output.csv

Location:

source-data container

Deliverable:

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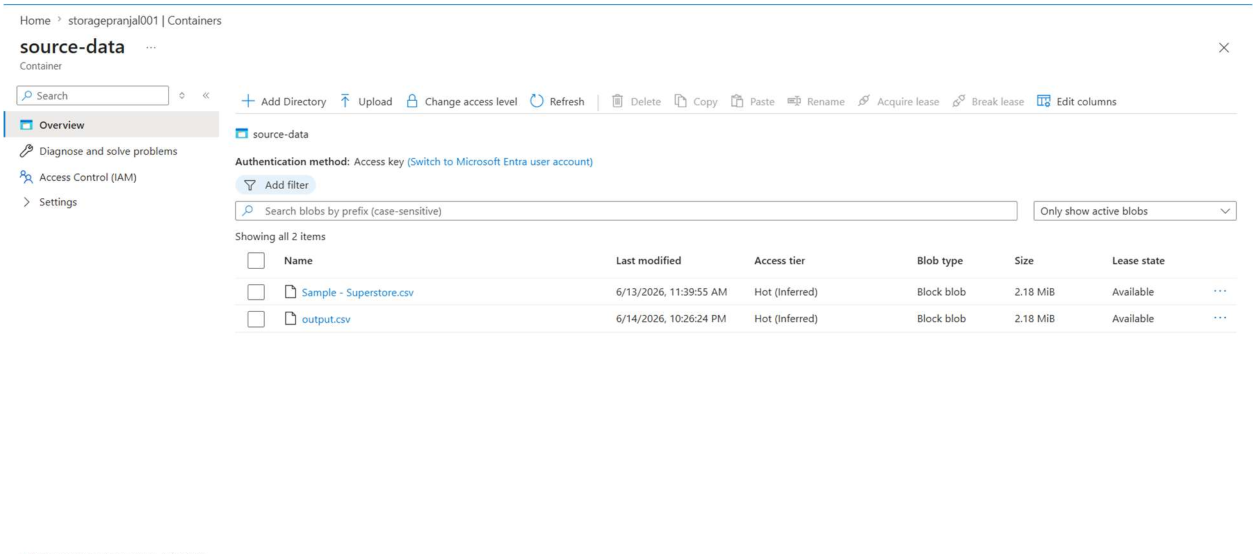
• Screenshot showing output.csv in source-data container



Output

Metadata Validation

The Get Metadata activity successfully validated:

- File Exists
- File Size
- Last Modified Date

Data Copy

The CSV file was copied successfully from:

source-data/Sample - Superstore.csv

to

destination-data/output.csv

Pipeline Status

Pipeline execution completed successfully.

Status: **Succeeded**

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Results

✓ Resource Group created successfully

✓ Storage Account created successfully

✓ CSV file uploaded successfully

✓ Linked Service configured successfully

✓ Source and Destination Datasets created successfully

✓ Metadata validation completed successfully

✓ Copy Data activity executed successfully

✓ Pipeline executed successfully

✓ Output file generated successfully

✓ Data copied to destination location

Conclusion

The project successfully demonstrates the implementation of a complete Azure Data Factory pipeline integrated with Azure Blob Storage. The source CSV file was validated using the Get Metadata activity and copied to a new destination location using the Copy Data activity. The pipeline executed successfully and produced the expected output file, fulfilling all project requirements.